



**STRUCTURAL DESIGN AND FEASIBILITY ASSESSMENT OF
LOCALLY SOURCED ALUMINUM AND COMPOSITE MATERIALS FOR
A CUBESAT STRUCTURAL ENGINEERING MODEL**

**A Project Report Submitted to the Department of Aerospace Engineering,
Faculty of Engineering, Lagos State University, in Partial Fulfillment of the
Requirements for the Award of the Bachelor of Engineering (B.Eng.) Degree
in Aerospace Engineering (First Semester Defense).**

July, 2026

CubeSat Structural & Vibration Analysis Report

1.0 Methodology & Setup

The FEA was conducted using Ansys Mechanical 2024 R1. The workflow utilizes a linked analysis approach:

1. **Static Structural:** Computes initial stresses and boundary conditions.
2. **Modal Analysis:** Determines the natural frequencies and mode shapes,
3. **Random Vibration:** Applies Power Spectral Density (PSD) base excitation to simulate launch loads.

2.0 Material & Mass Properties

The CubeSat primary structure is fabricated from locally sourced Aluminum 6063-T6. To account for internal electronics and payloads without explicitly modeling them, a dummy point mass of approximately 1.2282 kg was inserted and scoped to the mounting holes of the aluminum frame. This ensures the total mass and dynamic response accurately reflect the fully integrated satellite.

3.0 Boundary Conditions & The Saint-Venant Method

To accurately mirror the physical constraints of a P-POD deployer, Fixed Supports were applied to the base feet, while Frictionless Supports were applied to the sliding side rails. Additionally, bonded contacts were utilized at the structural interfaces to attach the dummy mass.

In finite element analysis, perfectly rigid boundaries (like the fixed support) and Multipoint Constraint (MPC) equations (used for the point mass) transfer massive inertial loads across infinitely sharp geometric edges. This mathematical behavior results in artificial "singularities" infinite stress spikes concentrated entirely at single nodes that do not represent physical reality. To resolve this and evaluate the true safety of the structure, the Saint-Venant Method was applied. By dropping localized probes one to two element lengths away from these contact boundaries, the localized mathematical anomalies were bypassed, allowing the capture of the true nominal equivalent (von Mises) and normal stresses propagating through the material. Furthermore, the Random Vibration analysis was configured to report 3-sigma (3σ) values directly, ensuring a 99.73% statistical probability for conservative margin of safety calculations.

The following boundary conditions and loads were applied across the analyses:

4.0 Setup & Boundary Condition Figures

Figure 1: Geometry (Global)

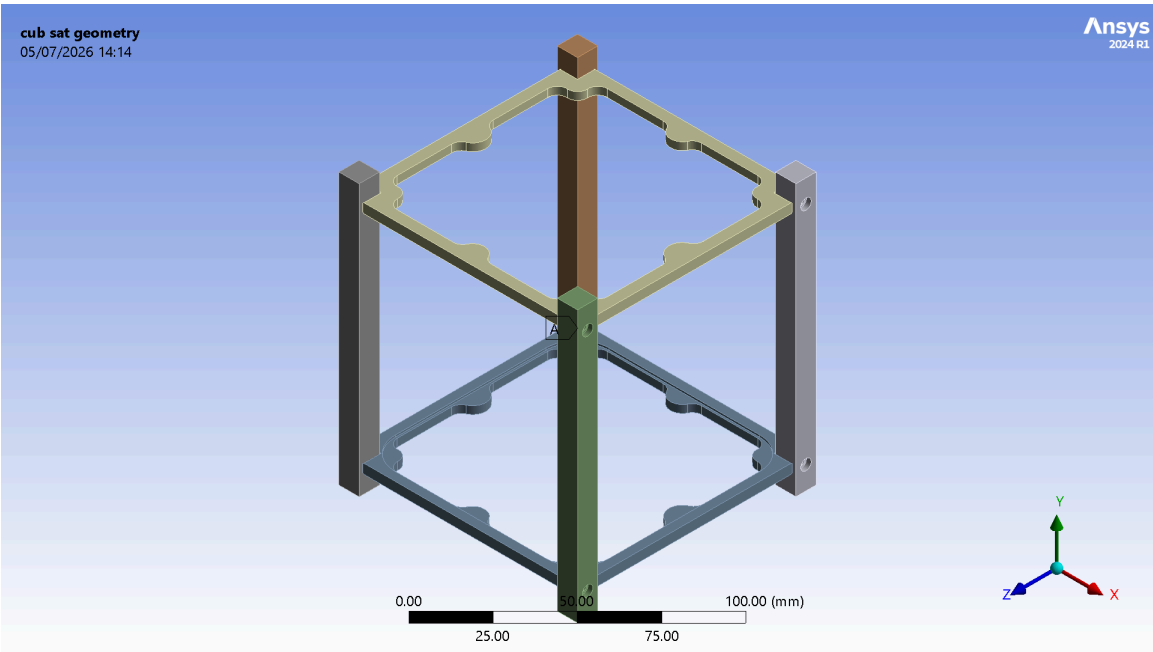


Figure 2: Bulkhead Coordinate System Z up (Global)

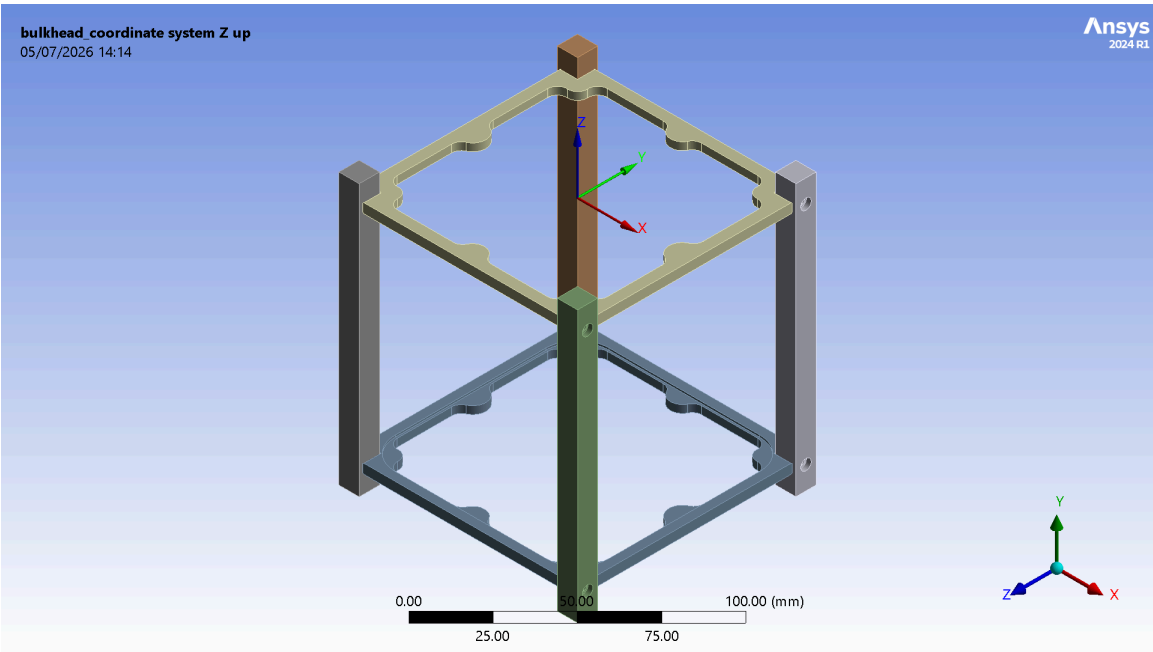


Figure 3: Acceleration boundary condition (Static Structural)

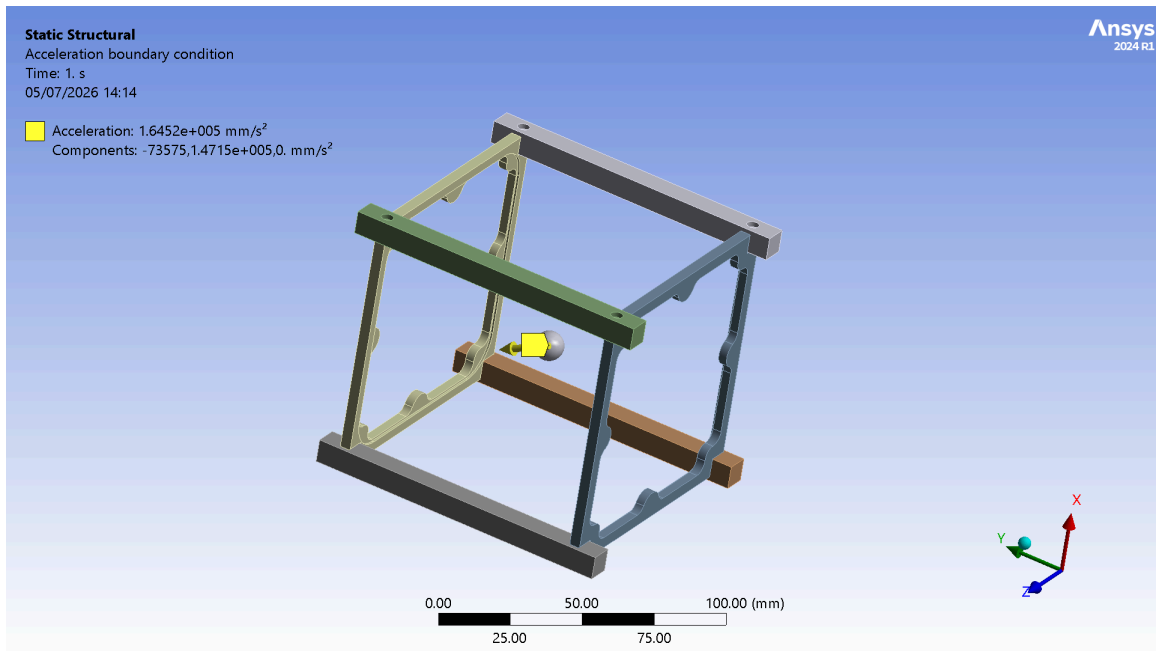


Figure 4: Fixed Support (Static Structural)

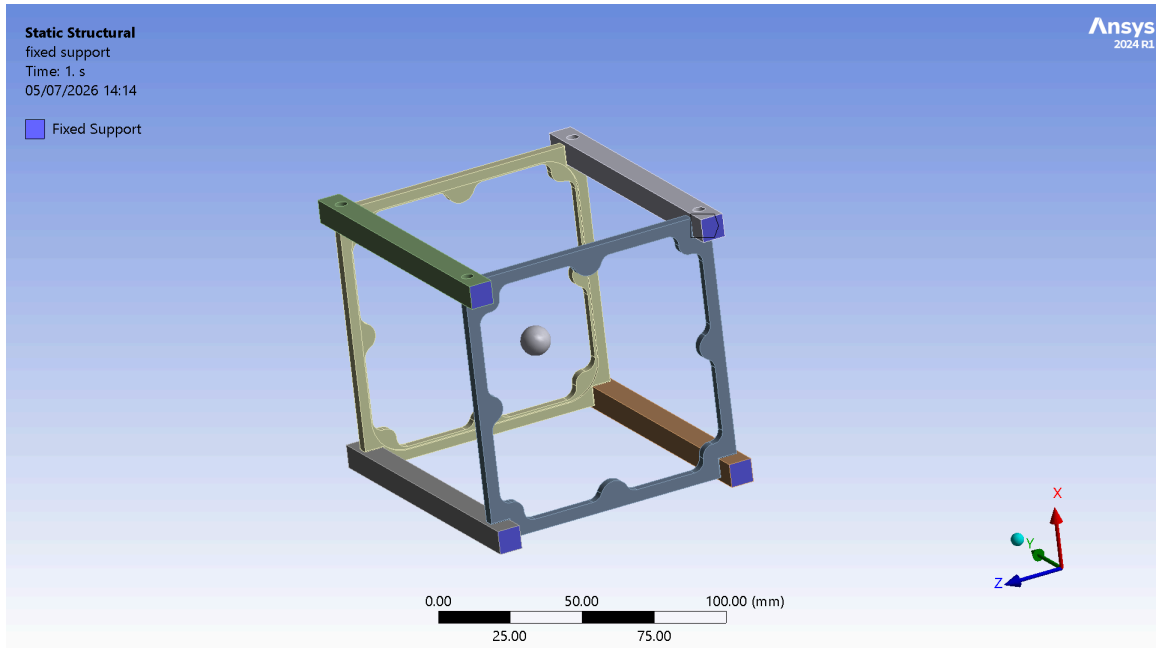
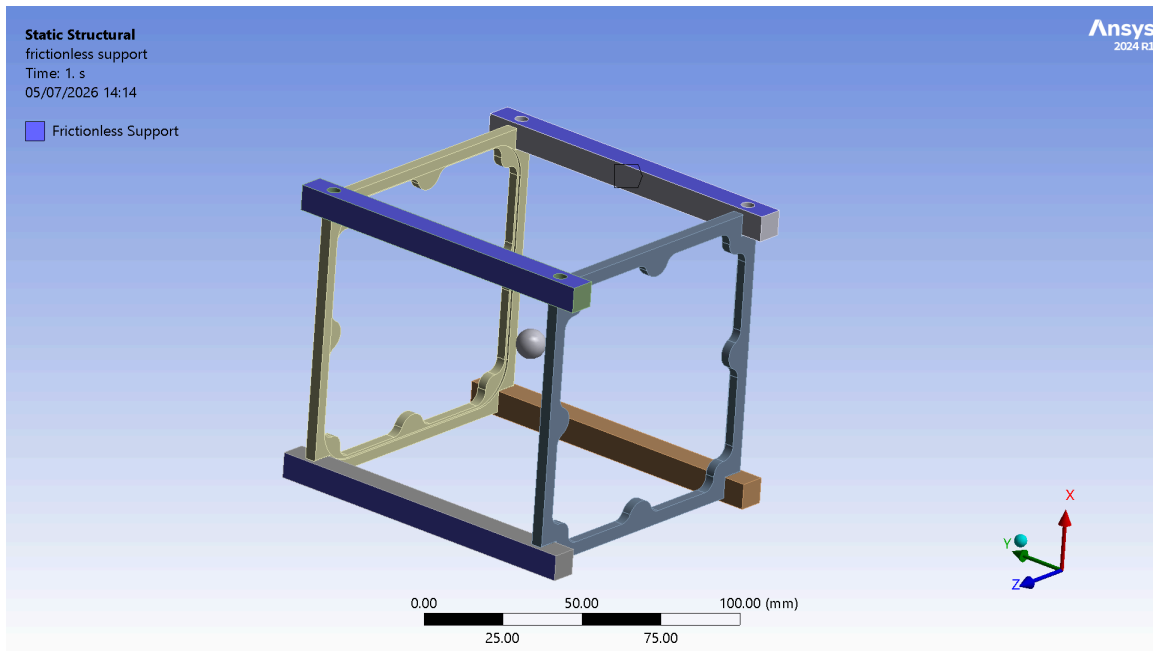


Figure 5: Frictionless Support (Static Structural)



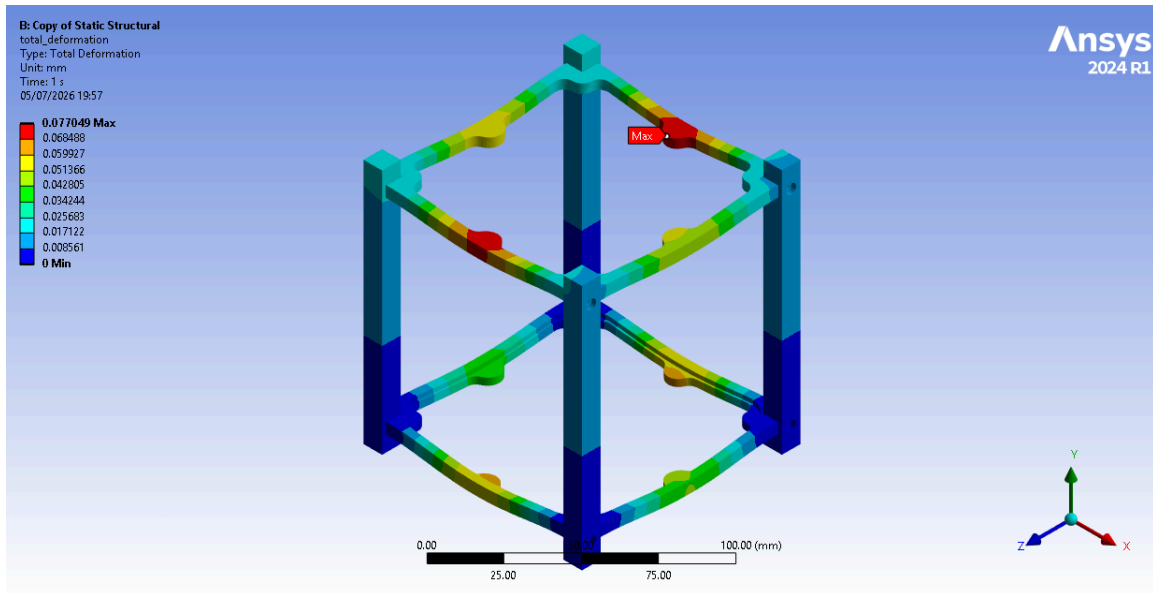
5.0 Results

5.1 Static Structural Analysis

The baseline static load verified the mesh and boundary condition setup. The maximum recorded Total Deformation was a negligible 0.11 mm. The maximum Equivalent Stress reached 0.78 MPa, establishing a clean, low-stress foundational matrix prior to dynamic loading.

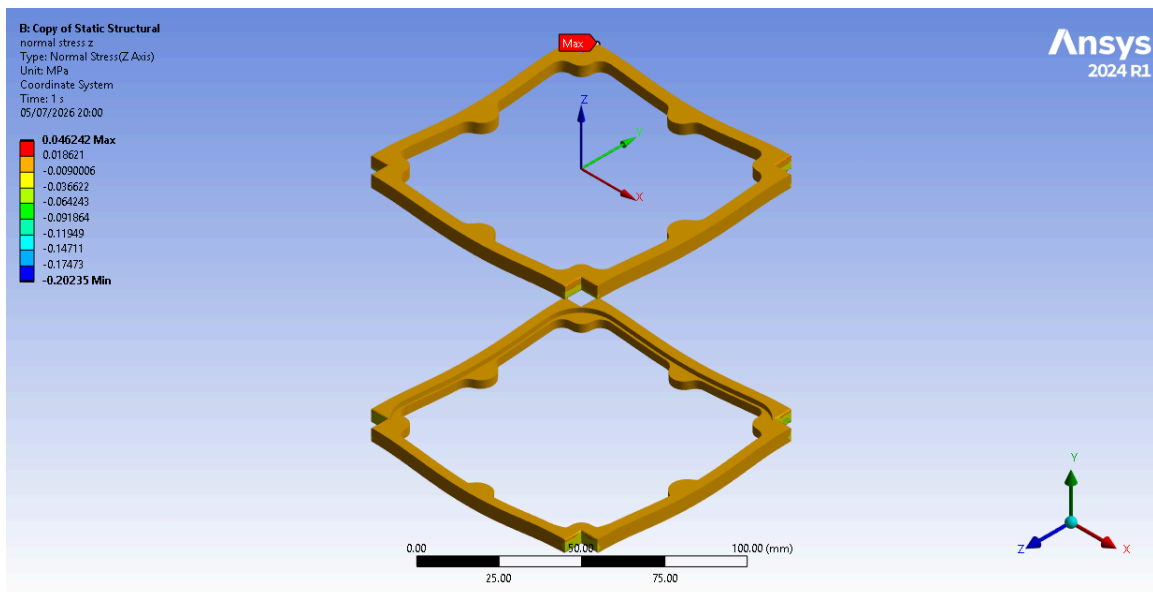
5.1.1 Result: Total Deformation

Maximum Value: 0.077 mm



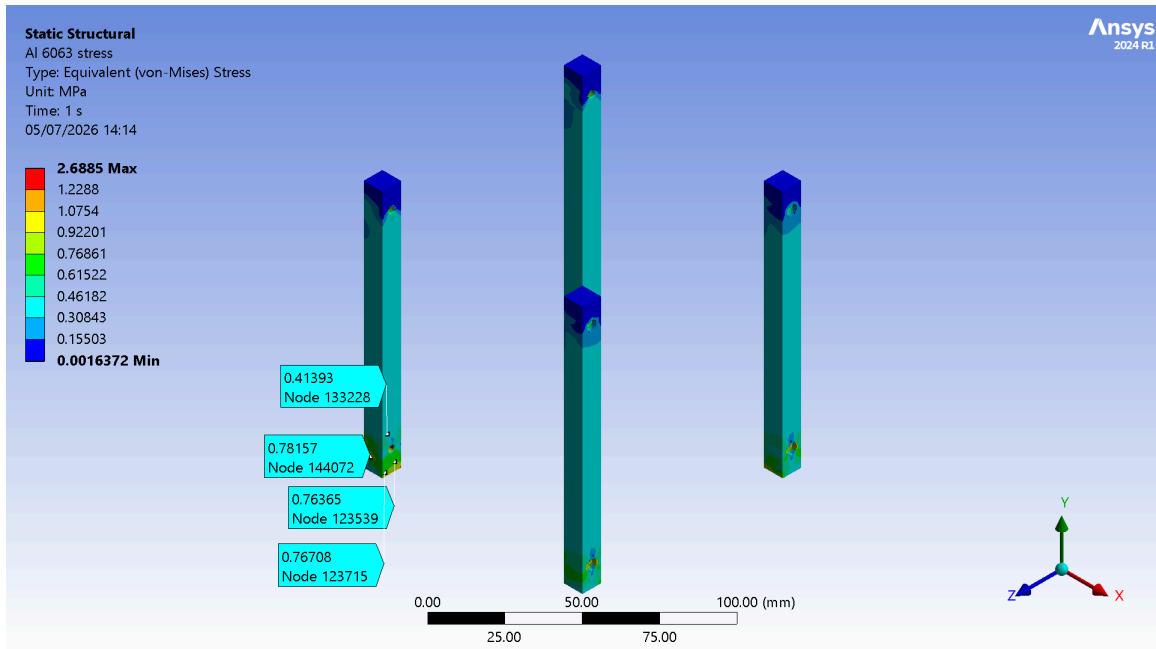
5.1.2 Result: Normal Stress Z axis

Maximum Value: 0.046



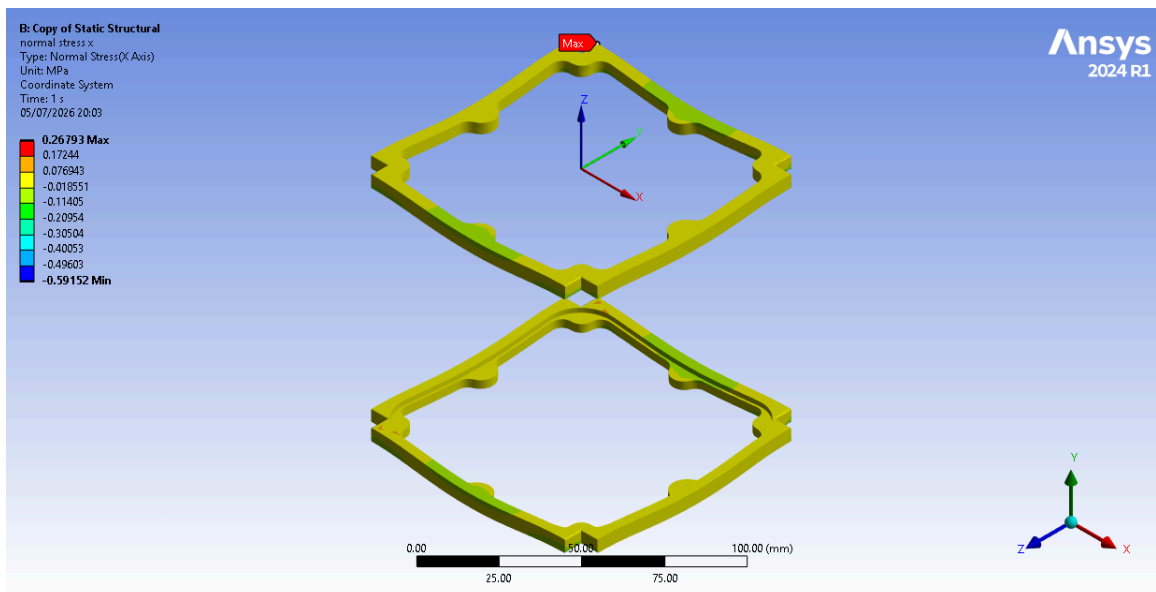
5.1.3 Result: Equivalent Stress AL 6063

Maximum Value: 0.78



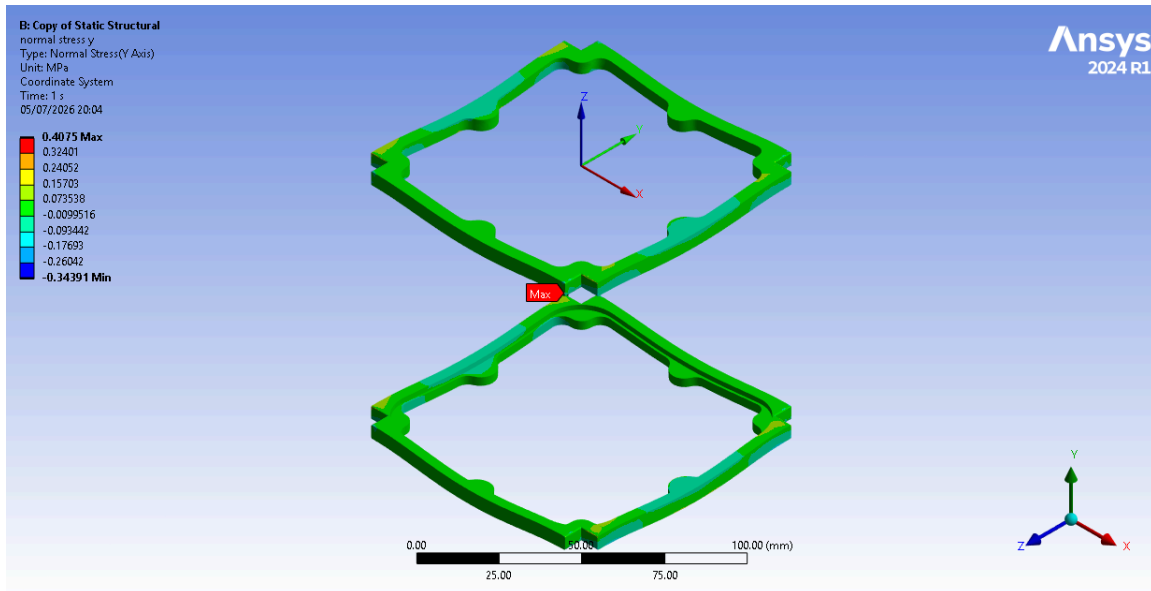
5.1.4 Result: Normal Stress x

Maximum Value: 0.2679



5.1.5 Result: Normal Stress 3

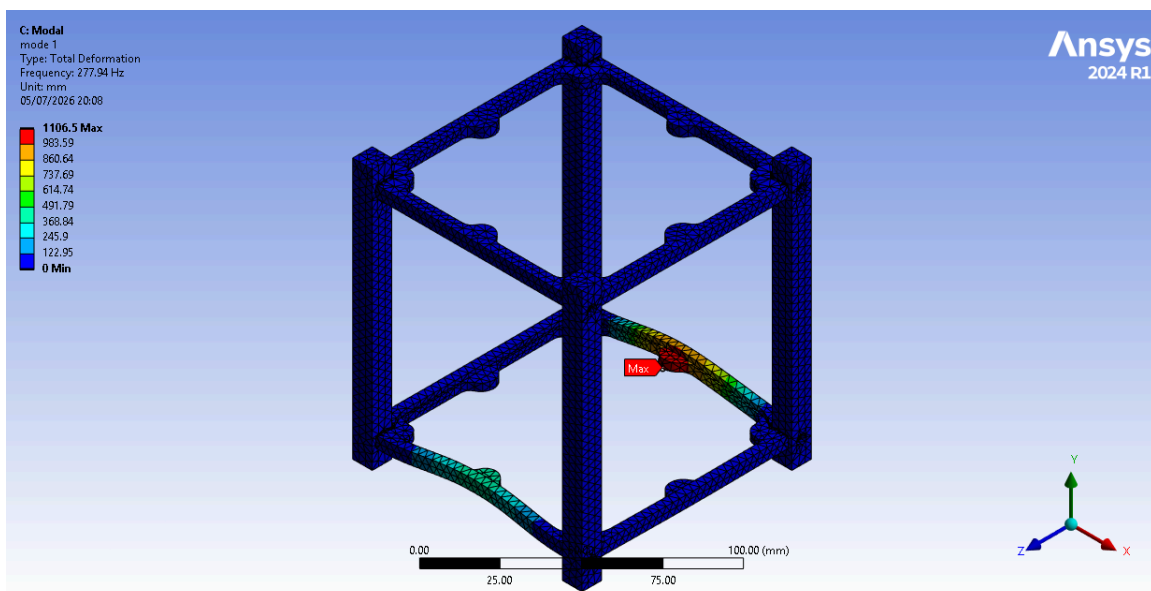
Maximum Value: 0.28



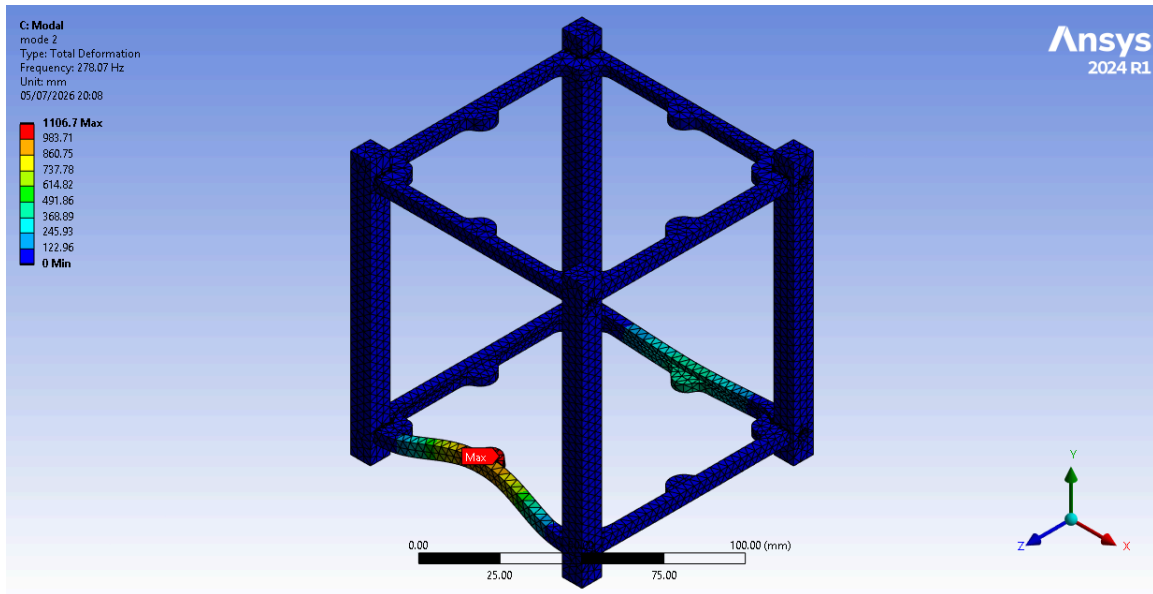
5.2 Modal Analysis

The modal solver extracted the first 10 natural frequencies of the system, determining how the structure naturally oscillates.

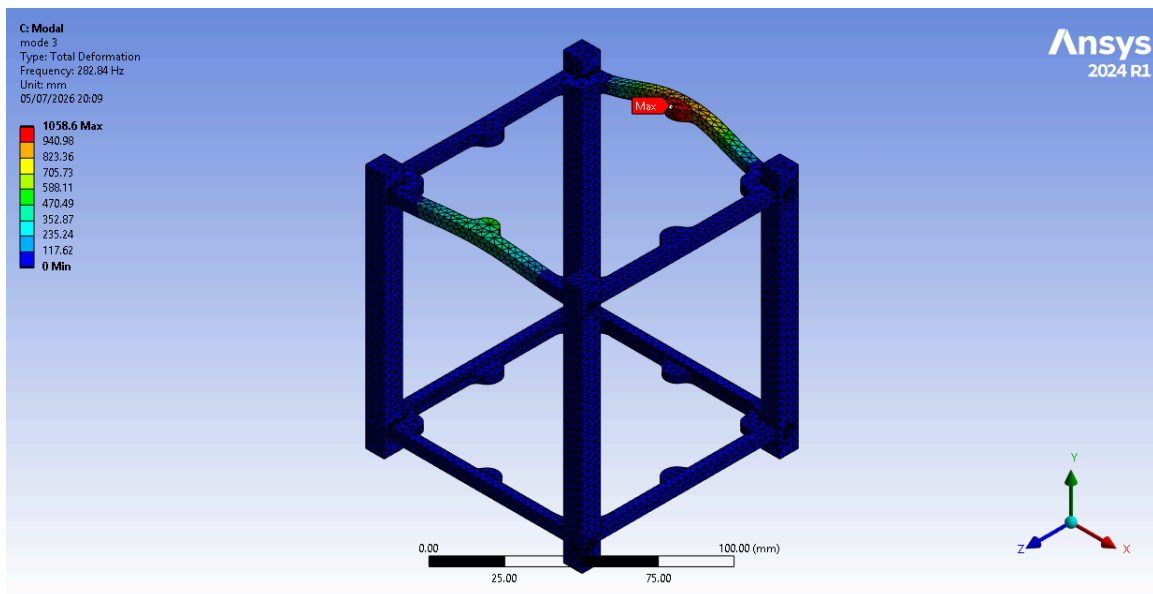
5.2.1 Result: Mode Shape 1



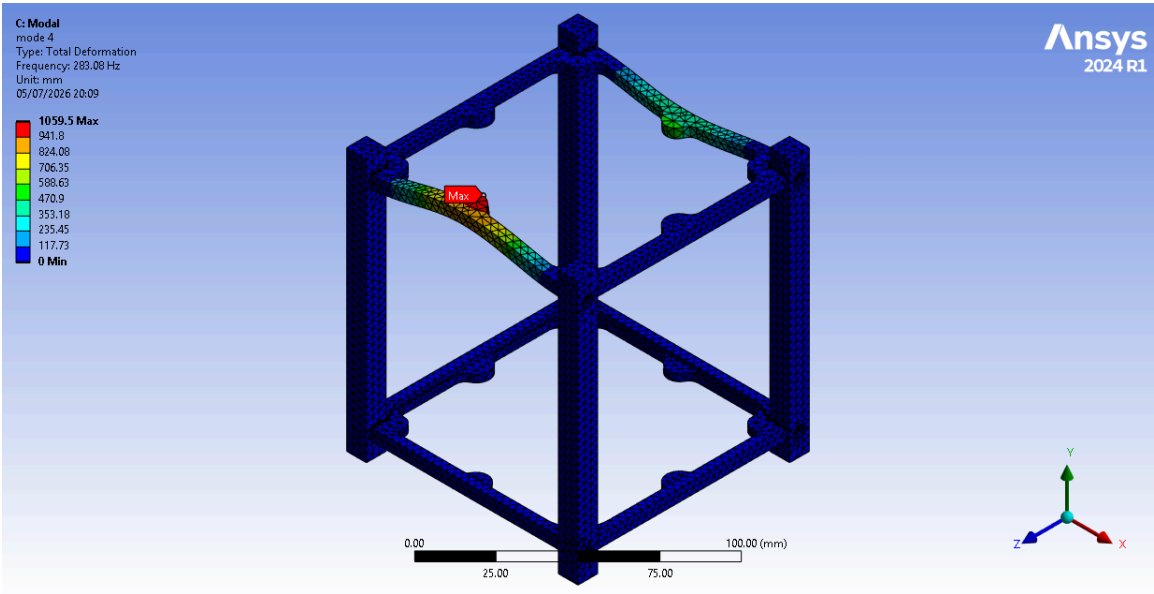
5.2.2 Result: Mode Shape 2



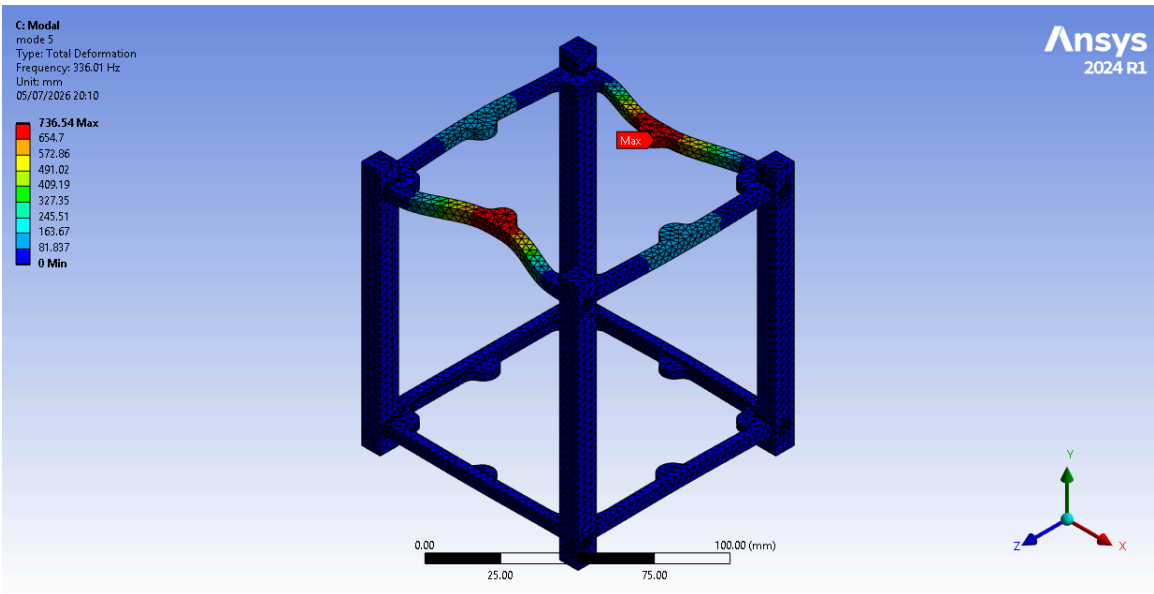
5.2.3 Result: Mode Shape 3



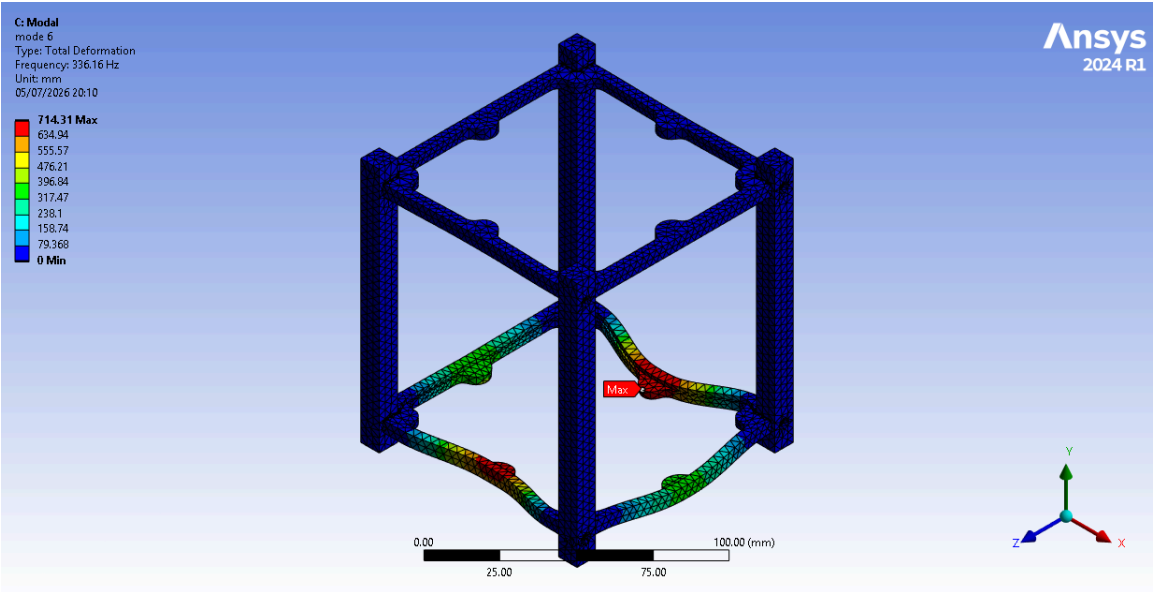
5.2.4 Result: Mode Shape 4



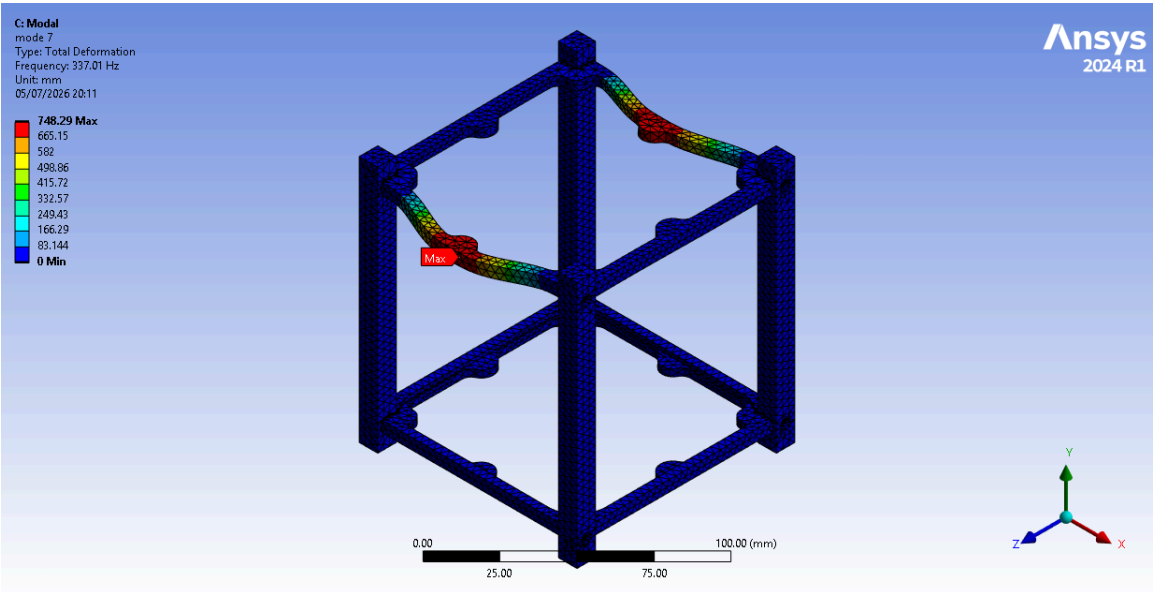
5.2.5 Result: Mode Shape 5



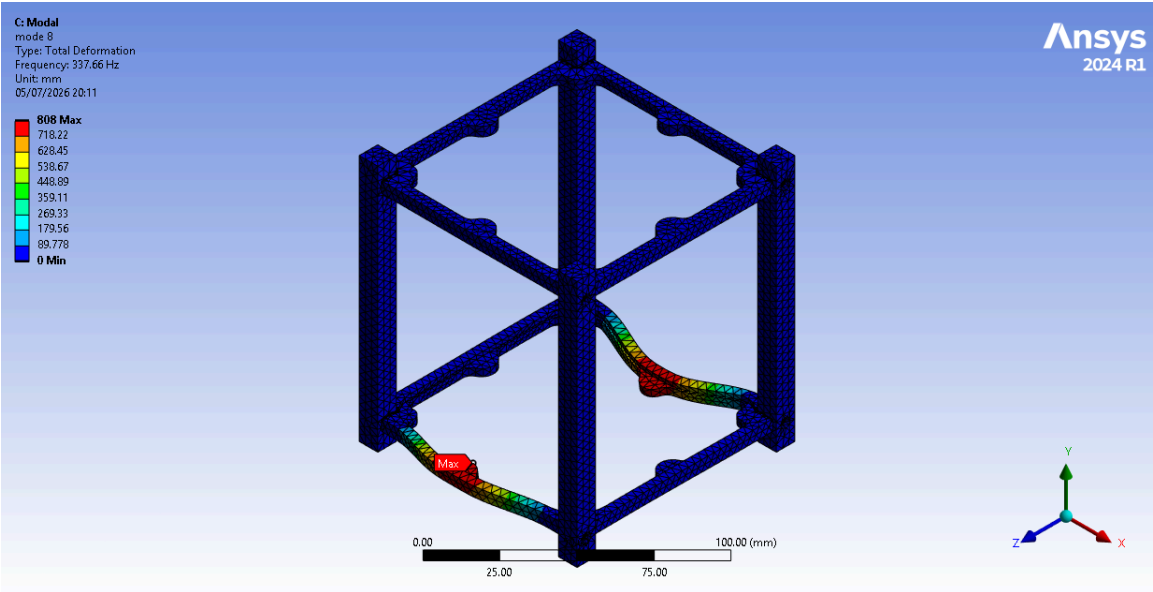
5.2.6 Result: Mode Shape 6



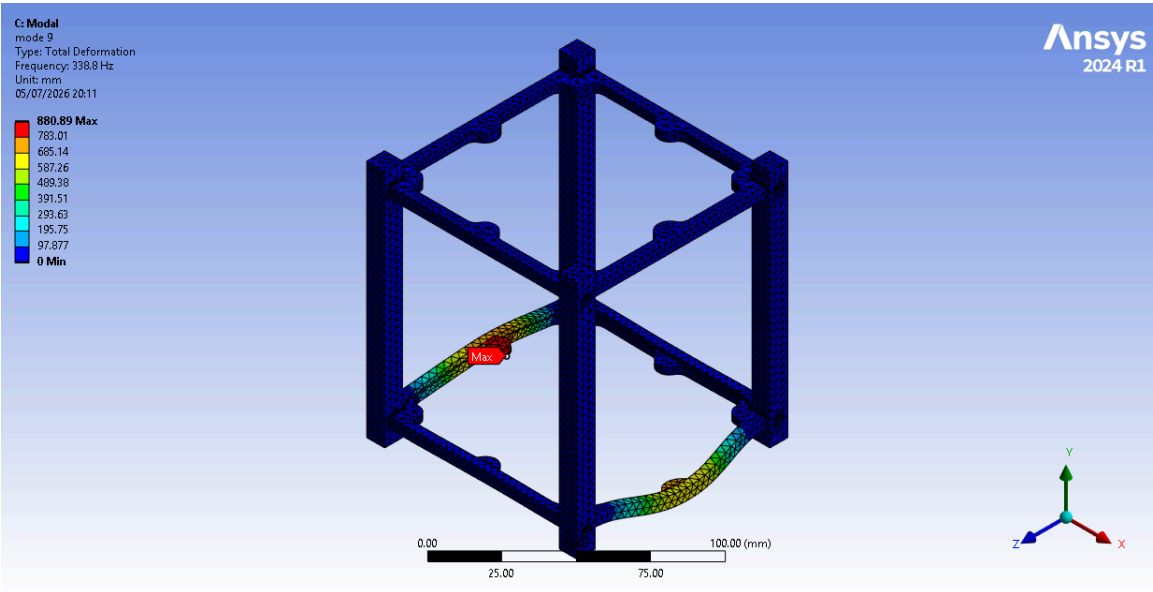
5.2.7 Result: Mode Shape 7



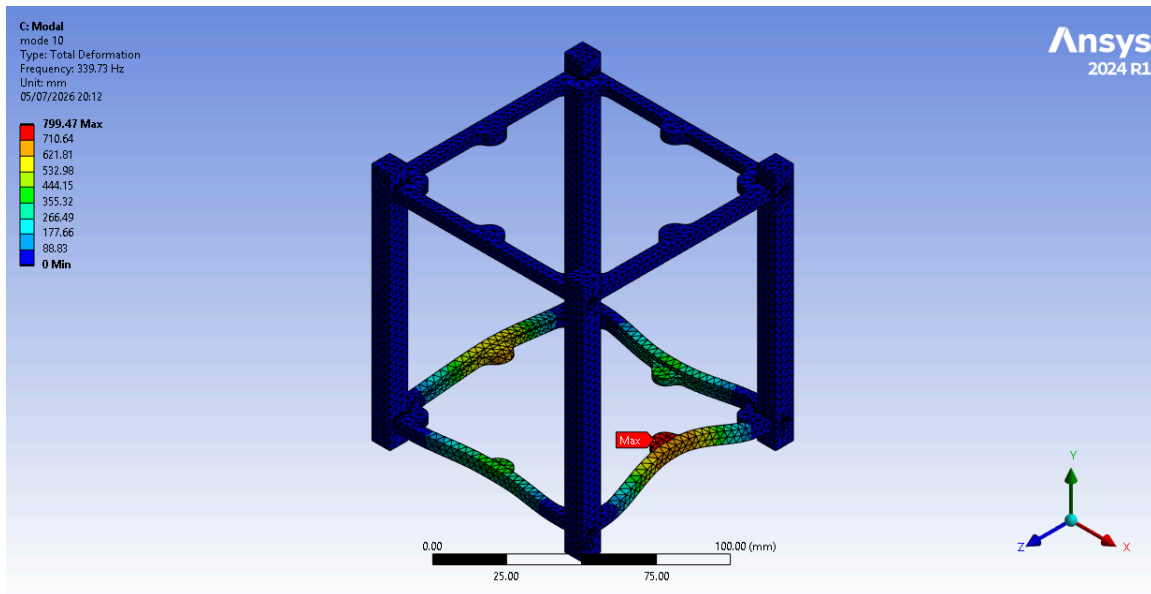
5.2.8 Result: Mode Shape 8



5.2.9 Result: Mode Shape 9



5.2.10 Result: Mode Shape 10



5.3 Random Vibration Analysis

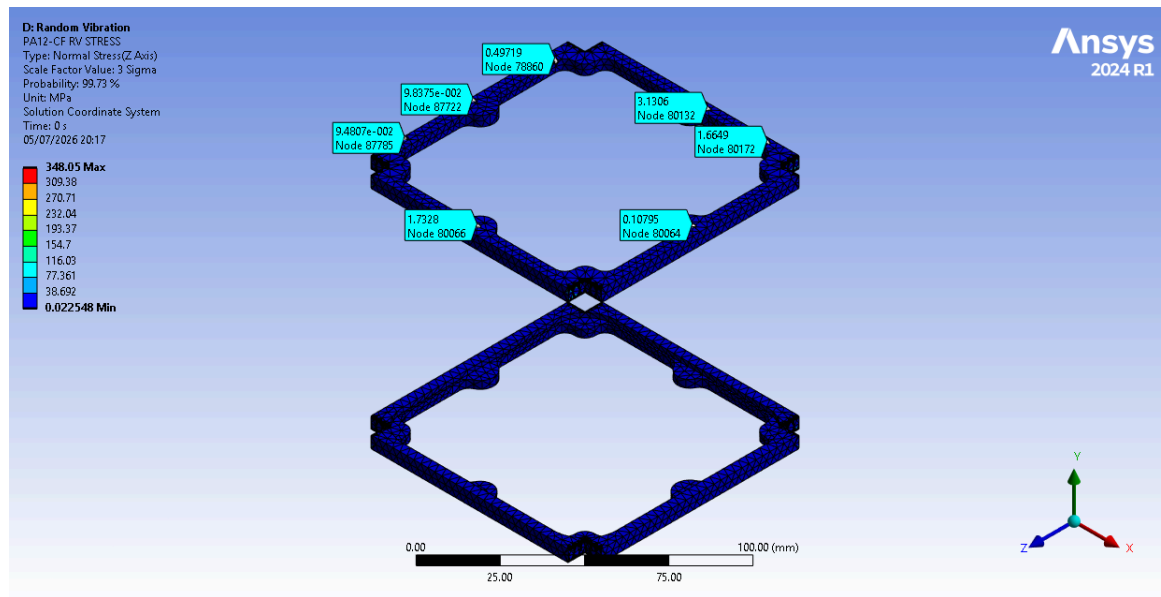
Following the application of the NASA GEVS random vibration PSD profile, the extreme unprobed contour peaks were mathematically confirmed as localized contact singularities. Utilizing the Saint-Venant probes, the true peak structural responses were recorded as follows:

- **Composite Panels (Normal Stress Z-Axis):** The maximum 3-Sigma out-of-plane stress was found to be 3.1306 MPa, ensuring layer adhesion limits are not breached.
- **Aluminum 6063 Frame (Equivalent von Mises Stress):** The maximum 3-Sigma physical stress on the primary rails was 71.726 MPa.

With the Aluminum 6063-T6 possessing a yield strength of 162.8 MPa, the peak load of 71.726 MPa guarantees a robust positive margin of safety, meaning the frame will not undergo ductile yielding during launch.

5.3.1 Result: Normal Stress

Maximum Value: 3.1306



5.3.2 Result: Equivalent Stress 4

Maximum Value: 71.726

